

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019**Brain Tumor Detection from Multimodal MRI Brain Images using
Pseudo Coloring Processes**T. Kalaiselvi^a, P. Kumarashankar^a, P. Sriramakrishnan^b, S. Karthigaiselvi^a^a The Gandhigram Rural Institute – Deemed University, Gandhigram, Tamilnadu - 62430, India^b Kalasalingam Academy of Research and Education (Deemed to be university), Krishnankoil, Tamil Nadu - 626126, India.

Abstract

In this work, we come up with a novel method to merge the color channels created from multi modal images to form a RGB image and then highlight the tumor in the MRI brain images, detect and extract the tumor region from the human head scan MRI images. This extraction process is done purely based on multimedia color channels and the basic characteristics of a MRI brain image. The original multi modal grayscale images (FLAIR, T2, T1c) are used to build the Red, Green and Blue channels respectively and then combined to form a whole RGB image using the pseudo coloring process. The Pseudo colored brain images are then analyzed for the presence of specific colors in the images which clearly identifies the tumor region in the whole brain image. The tumor and necrotic regions are shown in the RGB images in Golden Yellow color and White color respectively. Experimental study was done on BRATS high glioma imagesets. The main objective of this work is to detect the tumor in the human brain scans. A detailed study has to be done to detect the various regions of tumor. The results given by the proposed method reached good qualitative and quantitative level.

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1. Introduction

Magnetic Resonance Imaging (MRI) is a non-invasive technology that is used to produce three dimensional anatomical images. This is basically used for disease detection, diagnosis and treatment planning. This technique uses powerful magnets that produce a strong magnetic field because of which the protons in the body align with that magnetic field. A radio frequency current is then passed through the patient. This stimulates the protons and makes them spin out of equilibrium straining against the pull of the magnetic field. Once the current is off, the protons release the energy and realign with the magnetic field which are detected by the MRI sensors. The time taken by the protons to realign with the magnetic field, and the amount of energy released, will change depending on the environment and the chemical nature of the molecules which helps the doctors to identify the difference between the various types of tissues based on the magnetic properties. For MRI brain scans the image easily differentiates between the white and grey matter and is very useful to diagnose tumors in brain.

A brain tumor is an abnormal growth of cells in the brain [1]. Primary brain tumors originate from the brain. Metastatic brain tumors start as cancer in some other part of the body and spread to the brain. There is no specific reason that is known to cause brain tumors but they are treatable. The various subsections of brain tumors are: necrotic, active tumor, edema as given in Fig.1 and Fig.2. They have necrotises, which are dead cells of a tumor core. These necrotises are usually surrounded by active tumors. Edema, a form of fluid will be filled in-between and around these two regions [1] [2]. MRI is the most commonly used technique for brain tumor diagnostic process. A sample axial T2-weighted MRI brain tumor scan image along with the tumor subsections is shown in Fig.2.

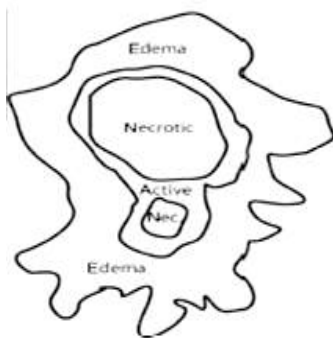


Fig. 1: Sub-Regions of Brain Tumor

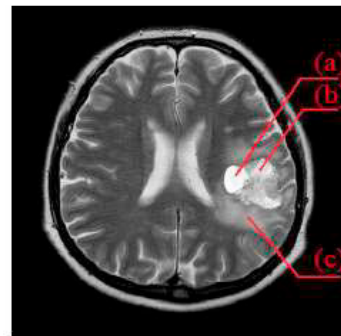


Fig. 2. T2-weighted image showing the brain tumor

(a) Tumor. (b) Necrotic tissue. (c) Edema

The third commonly used sequence is the Fluid Attenuated Inversion Recovery (FLAIR). FLAIR images are similar to T2-weighted images except that the TE and TR times are very long. Because of this, the abnormalities remain bright but normal CSF fluid is attenuated and made dark. This sequence is thus very sensitive to pathology and is very helpful to make the differentiation between CSF and an abnormality easier. When contrast agents are used for T1 images, the

contrast agents are injected during the scan they change the intensities by shortening T1. These contrast agents provide more brightness to T1 weighted images. They are more helpful in imaging the tumors [3].

Initial diagnosis, detection and segmentation of tumors from MRI images help to plan the treatment and track the progress of treatment; observe the growth or reduction in tumor size etc. There are various methods available for tumor detection and segmentation. Various techniques are available for segmentation including threshold based, region based, cluster based and classification techniques. Some of the very famous methods are k-means, Fuzzy C Means (FCM), support vector machines (SVM), artificial neural networks, etc., All of these methods use the basic tumor characteristics like intensity, structural, textural properties to detect the abnormalities. The accuracy is based on the knowledge that is used to define the tumor features and its boundaries [4].

There are various works done using coloring processes for medical images. One work is about coloring processes used for X-ray images. Various attributes in an X-ray image may not be clearly visible in grayscale whereas a colored X-ray image can easily show the attributes [5]. A work done by H Li et al., talks about a new approach for MRI brain image segmentation using pseudo coloring Non symmetry and Anti-packing model with Squares (NAMS). Sub patterns are used in this method to reduce redundancy. The pseudo colored images are then used for segmentation [6].

The proposed work generates a RGB image from the multi modal imagesets of the human head scans. This method uses the basic brain tumor characteristic of these images and multimedia color models to detect the tumors. The grayscale images are converted into RGB images and the tumor color characteristic is identified. Based on the color of the tumor the input intensity for the red, green and blue channels are determined based on which the tumorous region is identified and compared against the ground truth and documented the results. The experimental study is conducted over a random set of high glioma imagesets of BRATS 2012. We have used three types of imagesets FLAIR, T2, T1c as the red, green and blue channels respectively.

2. Multimedia Color Models

Color is a very essential component of visual multimedia. There is basically a very big difference in how humans process the images and color compared with computers. The computers always look at an image as a matrix of pixels with varying intensities. The human eye reasons the image that it captures and can easily do the processing based on the knowledge. But a computer has to be trained enough to understand the various types of images and their characteristics. A color model is a system of creating a complete range of colors from a set of primary colors.

There are two basic types of color models, additive and subtractive. The additive color model uses light for computer display whereas the subtractive model is basically used in printers. This model uses red, green and blue colors. The various combinations of these three colors give a wide spectrum of colors that is used in computer display. This color model uses light and the absence of light is conceived to be black. The model is also called as RGB model (Figure 3). In order to create a color three colored light beams should be super imposed. With no intensity, the color formed is black and with full intensity the color formed is white. With the use of 24-bit displays, there could be approximately 16.78 million colors that can be created.

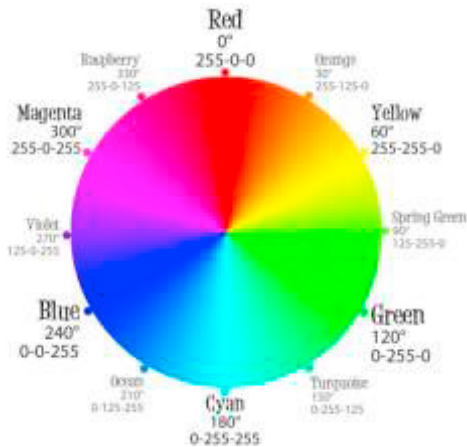


Fig.3. RGB Color Spectrum

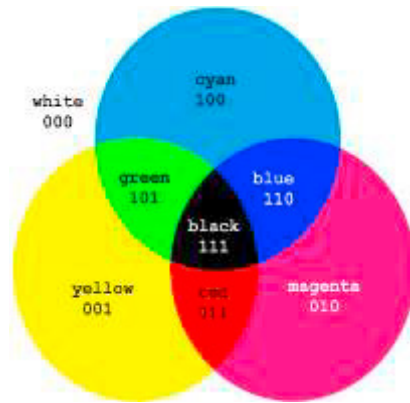


Fig. 4. CMYK Color Spectrum

The subtractive color model uses the cyan, magenta and yellow as primary colors to create a reflection of the colors. When two subtractive primary colors are combined, an additive primary color is formed. This color primarily works by partially or completely masking colors in a white background. The model is also called as CMYK model (Figure 4). The K in the name denotes ‘key’ as the CMY printing plates are carefully aligned with that of the key of the black key plate. The combination of the three primary colors cannot create a proper black color; hence the black color is also used in this model.

The RGB and CMYK models are given in Figure 3 and Figure 4 respectively. There are other color models like Bitmap, HSV, Grayscale, Duotone, halftone, HSL, Indexed Color, LAB, and NCS etc. A bitmap means a map of bits. This is a very straight forward model wherein only one bit is used to denote the color, either ‘1’ for white or ‘0’ for black. For image processing, we use these bitmap images for creating masks and also for functions like shape analysis.

HSV stands for hue, saturation and value. This model is used to depict relationships between various colors on top of the RGB model. If the HSV model is considered as a wheel, then hue is the angle that is expressed as a number from 0 to 360 degrees representing hues of red (starts at 0), yellow (starts at 60), green (starts at 120), cyan (starts at 180), blue (starts at 240), and magenta (starts at 300). Saturation is the amount of gray in the image represented as percentage (0% to 100%) in the color. Value describes the brightness or intensity of the color from 0% to 100%.

The grayscale model contains a range of gray values starting from black to white. This basically contains only the intensity information of the pixels. The images formed using this model are basically black and white and there will be varying shades of gray in the image. These images are monochromatic.

3. Proposed Method

The proposed work uses the multimodal images of random high glioma BRATS 2012 dataset. We have used the Axial FLAIR T2 and T1c imagesets for this proposed work. The gray colored imagesets are given as input to this method. The gray color images are mapped into red, green and blue channels and an RGB image is created. The

FLAIR image is used for red channel, T2 is used for green channel and T1c is used for blue channel respectively. The intensity of these grayscale images are used to specify the same intensity of respective colors in the method. The tumor region and necrotic region are clearly visible from the output RGB imageset. The tumor region is displayed with golden yellow color and the necrotic region is shown as white color.

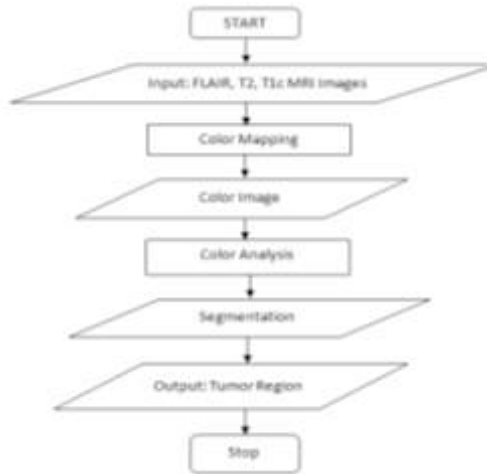


Fig. 5: Flowchart of the proposed method

The proposed work is described in the below steps (Flowchart in Fig 5):

Input: MRI FLAIR, T2 and T1c grayscale imagesets

Step 1: Pseudo Colouring

Step 2: Colour Analysis

Step 3: Segmentation & Detection

Output: Tumor region

3.1 Pseudo Coloring

The grayscale MRI FLAIR, T2, T1c images are converted to RGB color model using the pseudo coloring process. Each of these is an 8-bit image that could contain gray shades varying from 0 to 255. Then the intensity values of the FLAIR images are taken as the intensity of red channel, of the T2 images are taken as the green channel, and that of the T1c images are taken as the blue channel. `cat` and `rgb2gray` functions in Matlab are used to convert grayscale to RGB and vice versa[8]. These red, green and blue channels are combined to form a RGB image. With the varying intensities of various multi modal images, the RGB image created contains a mix of a range of colors. The input images (FLAIR, T2, T1c) are and the respective RGB image created are shown in Fig 6.

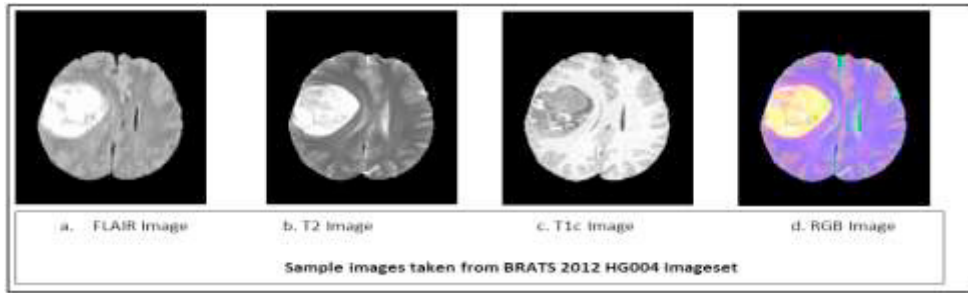


Fig. 6. Sample input images from BRATS 2012 HG004 Imageset and the resultant RGB image

3.2. Colour Analysis

Once the color image is obtained, the output RGB image shows different parts of the brain are displayed in different colors. The analysis of colors show that the tumor region is shown as golden yellow, necrotic region in white color, CSF region in green color, grey matter in blue color. Golden Yellow is a color halfway between amber and yellow.

$$RGB (> 150, > 150, < 175) \quad (1)$$

$$RGB (> 225, > 225, > 225) \quad (2)$$

$$HSV (52.5, 100, 100) \quad (3)$$

$$HSV (0, 0, 100) \quad (4)$$

$$CMYK (0, 16, 100, 0) \quad (5)$$

$$CMYK (0, 0, 0, 0) \quad (6)$$

Golden Yellow color contains high value for red and green and less value blue. To cover up the nearest shades of golden yellow color we use the value as depicted in equation (1). There is a dominance of red and green for the golden yellow color which is seen in equation (1). For white, all the three colors have higher values. In the HSV model the brightness is about 100% for the golden yellow color. The sample RGB image and the images obtained in subsequent steps are given in the Figure 7.

The RGB image clearly shows the tumor with the above colors of golden yellow and White. The whole tumor and necrotic regions are shown in golden yellow and white colors respectively. The golden yellow colored pixels in the image are denoted as $I(GY)$ and white shades are denoted as $I(W)$.

3.3. Segmentation & Detection

After the RGB images are created for all the imagesets, based on the color analysis, all the golden yellow shades $I(GY)$ and white shades $I(W)$ are marked as pure white. This is done by adjusting the red, green and blue channels to the highest intensity possible (255).

$$I(GY) = 255 \quad (7)$$

$$I(W) = 255 \quad (8)$$

The resultant images will contain the tumor marked as pure white with the above equations 7 and 8. Once these images are adjusted for this intensity, then from the entire image only the high intensity pixels are extracted which are bright and are marked with high intensity in the above step and the remaining pixels are marked with zero intensity.

$$I(BC) = I(GY) \cup I(W) \quad (9)$$

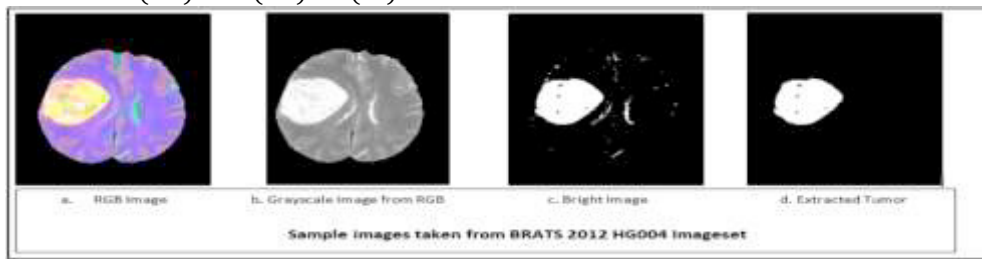


Fig. 7. Resultant RGB image from BRATS 2012 HG004 Imageset and the corresponding images in every step of proposed method

Imageset – Slice No	FLAIR	T2	T1C	RGB	Bright	Tumor
HG001 – 1093						
HG0002 – 1085						
HG0003 – 1114						
HG0004 – 1104						
HG0008 – 1083						
HG0009 – 1115						
HG0014 – 1103						
HG0022 – 1069						

Fig. 8. Input images and subsequent images obtained from the method.

Whereas $I(G)$ is also marked as White in equation 7. From the resultant image, it is seen that the tumors region is completely bright and is clearly visible in the resultant images. In the next step it will help to identify and extract the tumorous region. The images are again converted to grayscale.

$$I(G) = I(BC) \quad (10)$$

From the extracted images, the tumorous region is shown big and bright [7]. To extract the tumor from the images, the biggest bright portion from the images is chosen. The input images and the subsequent images from every step of this method are given in Figure 8. The big bright (BBC) portion in the images are extracted and marked as tumor. The big bright interconnected portion of the image is taken as tumor and further analysed with ground truth images for verification and validation as discussed in the next section.

$$I(BBC) = I(G)i \quad (11)$$

Where $i > 255$, and $I(BBC)$ is the biggest interconnected region in the image $I(G)$. $I(BBC)$ contains only the biggest interconnected bright region and the rest of the pixels are simply marked with 0 intensity.

4. Results and Discussion

The experimental study is done for randomly selected tumor volumes of BRATS2012 imagesets. Qualitative analysis was done manually for the results and compared with ground truth images. The sample input images and the corresponding output images are given in Figure 8. Column 1, 2, 3 show the original images FLAIR, T2, T1c respectively. Column 4 shows the RGB image, Column 5 shows the RGB image with tumor region set to the highest intensity for all three RGB channels and converted to grayscale, hence showing the tumor region as White. Column 6 shows the extracted tumor. The RGB image shows the tumor region as Yellow and the necrotic region as White. In this proposed work, our interest is only to detect tumors and hence there is no differentiation study done in detail to extract the different regions of tumor. Only the entire tumor region is highlighted in the resultant images.

For quantitative analysis, accuracy (A) and similarity index (SI) are used as given below.

$$Accuracy(A) = (TP + TN) / (TP + TN + FP + FN) \quad (12)$$

$$Similarity Index(SI) = 2TP / (2TP + FP + FN) \quad (13)$$

The values of the true positive (TP), true negative (TN) false positive (FP), and false negative (FN) are calculated as given in Table 1.

Table 1: For Outcomes

Manual Segmentation	Proposed Result	
	Tumor	Non-Tumor
Tumor	TP	FN
Non-Tumor	FP	TN

Accuracy (A) and Similarity Index (SI) have values between 0 and 1. Eleven tumorous volumes were selected for quantitative validation and their clinical details are given in Table 2. On an average, the proposed work resulted in 99 % accuracy (A) and 83% SI.

Table 2: Performance of Proposed Method in terms of Accuracy (A) and SI to tumor region detection

Imageset	Sum TP	Sum TN	Sum FP	Sum FN	Overall Accuracy	Overall Similarity
HG0001	83766	5967487	6254	25053	0.99485299	0.842551009
HG0002	40509	6023713	10146	8192	0.996985151	0.815431378

HG0003	135420	6516627	20600	18169	0.994205639	0.874780772
HG0004	90490	5967200	8584	16286	0.995911261	0.879183872
HG0008	163509	5863697	18712	36642	0.990899555	0.855235216
HG0009	169125	5847111	15865	50459	0.989096039	0.836064601
HG0014	92236	6543542	5902	49136	0.991774098	0.770205837
HG0022	55767	8633285	10579	28869	0.995480552	0.738723821

The proposed work is a simple method that works on the concept of pseudo coloring and basic color models. The method is able to detect tumors with utmost accuracy in most of the imagesets that were used for the analysis. This method could give more accurate results when used along with classification techniques. It could be used to identify new features that could help in segmentation and classification of tumors. The proposed work is limited to detecting single presence of tumors in MRI head scans. Multiple tumors cannot be detected using this work. Any future work could improvise on this and can be aimed to develop advanced methods on top of this to detect multiple tumors as well.

5. Conclusion

The proposed work is a novel method to detect brain tumors using color components from the MRI human head scans. The method uses multi modal imagesets for pseudo coloring and analysis. The method multimedia based color models and the output is analysed for the presence of tumor. Visual analysis shows the presence of tumor and its sub regions. The study has to be furthered to specifically segregate the various regions of tumor. The quantitative analysis shows good results with high values of accuracy and similarity. The proposed work could be refined and fine-tuned by combining with additional features to give more robust and accurate results.

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